

01 SCOPE & THE FLUID-MOSAIC MEMBRANE

Biochemistry = chemistry of life processes. In veterinary practice it supplies the **diagnostic** (serum enzymes, BUN), **nutritional** (limiting amino acid, EFA), **therapeutic** (fluid therapy, PGF_{2α}) and **toxicological** (xenobiotic, P450) framework.

Fluid mosaic model — Singer & Nicolson, 1972. A fluid lipid bilayer in which globular proteins float like icebergs. Bilayer = amphipathic phospholipids (polar head out, acyl tails in) + cholesterol + glycolipids. Proteins are **integral/intrinsic** (span bilayer, need detergent, e.g. Na⁺/K⁺-ATPase, glycophorin) or **peripheral/extrinsic** (surface, salt extractable, e.g. spectrin, cyt c).

7–10

nm thickness

18%

protein — myelin (lowest)

76%protein — inner mito
(highest)**50:50**

lipid:protein, RBC ghost

- **Asymmetry:** outer leaflet = phosphatidylcholine, sphingomyelin, **all glycolipids & glycoproteins**; inner leaflet = phosphatidylethanolamine, phosphatidylserine, PIP₂. **PS flipped outward = "eat-me" apoptosis signal** (basis of Annexin-V assay).
- **Movement of lipids:** lateral diffusion & flexion are **fast**; transverse flip-flop is **very slow** and needs enzymes — **flippase** (inward, ATP), **floppase** (outward, ATP), **scramblase** (both ways, Ca²⁺-activated, no ATP).
- **Fluidity** ↑ by unsaturated & short-chain fatty acids and ↑ temperature; ↓ by saturated/long chains. **Cholesterol is a fluidity buffer** — below transition temp it prevents crystalline packing (↑fluidity); above it restrains chains (↓fluidity).
- **Functions:** selective permeability, compartmentation, electrical excitability, receptor & signal transduction, cell recognition (glycocalyx), antigenicity (blood groups), cell adhesion, energy transduction (mitochondrial inner membrane).

Trap: the bilayer is freely permeable to **small uncharged** molecules (O₂, CO₂, N₂, urea, ethanol, steroids). It is essentially impermeable to **ions and to glucose** — even Na⁺ and K⁺, being small, are blocked because of their large hydration shells.

02 TRANSPORT ACROSS MEMBRANES — THE MASTER MAP

MEMBRANE TRANSPORT

PASSIVE — down gradient, no ATP

Simple diffusion

no carrier · non-saturable
linear with gradient
O₂, CO₂, urea, steroids
+ aquaporins for H₂O

Facilitated diffusion

carrier · saturable (V_{max})
specific · inhibitable
GLUT 1–5, urea & anion exchanger (band 3)

ACTIVE — uphill, energy needed

Primary active

ATP hydrolysed directly
Na⁺/K⁺-ATPase (3Na out:2K in)
Ca²⁺-ATPase, H⁺/K⁺-ATPase
P-, F-, V-, ABC types

Secondary active

rides the Na⁺ gradient
symport: SGLT-1 (2Na:1glc)
antiport: Na⁺/Ca²⁺, Na⁺/H⁺
stops if ATPase poisoned

Uniport one solute, one way — GLUT1

Symport two solutes, same way — SGLT-1

Antiport two solutes, opposite — Cl⁻/HCO₃⁻

Bulk endo/exo/pinocytosis, phagocytosis

Ionophores valinomycin (K⁺ carrier) · gramicidin (channel)

Most-asked: Na⁺/K⁺-ATPase — electrogenic (net +1 out), consumes **≈30 % of basal ATP**, inhibited by **ouabain / digitalis / cardiac glycosides** and by K⁺-free medium; it creates the Na⁺ gradient that powers **all** secondary active transport, and maintains cell volume against Donnan swelling.

03 GLUT FAMILY — ONE LINE EACH

CARRIER	SITE	POINT THAT SCORES
GLUT-1	RBC, brain, BBB	Low K _m ≈ 1 mM — basal uptake
GLUT-2	Liver, β-cell, kidney, gut basolateral	High K_m ≈ 15–20 mM = the glucose sensor ; bidirectional
GLUT-3	Neurons, placenta	Lowest K _m — neuron gets first claim
GLUT-4	Skeletal & cardiac muscle, adipose	Only insulin-dependent one — vesicle recruitment
GLUT-5	Intestinal brush border, sperm	Fructose transporter
SGLT-1/2	Gut / renal PCT	Secondary active, Na ⁺ -coupled; phlorizin inhibits

04 GIBBS-DONNAN MEMBRANE EQUILIBRIUM

Statement. When a **non-diffusible charged ion** (plasma protein anion, Pr⁻) is confined to one side of a semipermeable membrane, the diffusible ions redistribute **unequally** so that the **product of diffusible ion concentrations is equal on both sides**, while electroneutrality holds on each side.

$$[\text{Na}^+]_i \times [\text{Cl}^-]_i = [\text{Na}^+]_o \times [\text{Cl}^-]_o \cdot \text{Donnan ratio } r = [\text{Na}^+]_i / [\text{Na}^+]_o = [\text{Cl}^-]_o / [\text{Cl}^-]_i \approx 0.95$$

- **Three consequences:** ① the protein side holds **more diffusible cations, fewer anions**; ② total osmotically active particles are higher on the protein side → water pulled in (**Donnan effect adds ≈ 1/3 extra to colloid osmotic pressure**); ③ a **Donnan potential** develops across the membrane.
- **Veterinary significance:** plasma vs interstitial fluid ion split; contribution to resting membrane potential; the **chloride shift (Hamburger phenomenon)** in RBC; oedema of hypoproteinaemia (parasitism, PLE, liver failure) — falling albumin collapses both oncotic and Donnan pressure; behaviour of colloids (dextran, plasma) in fluid therapy; ultrafiltration in the glomerulus.
- **Double Donnan / pump-leak:** if Donnan forces acted alone the cell would swell and burst. Na^+/K^+ -ATPase keeps Na^+ functionally impermeant outside, opposing it — cell volume is a **steady state, not equilibrium**.

05 DISSOCIATION OF ACIDS · PH · BUFFERS · HENDERSON-HASSELBALCH

Brønsted–Lowry: acid = proton donor, base = proton acceptor. Strength of a weak acid is its pK_a = pH at which it is exactly **half dissociated**. Lower pK_a = stronger acid. $\text{pH} = -\log[\text{H}^+]$; blood $[\text{H}^+] \approx 40 \text{ nmol/L}$.

Buffer = weak acid + its conjugate base; resists pH change on adding acid/alkali. **Buffer capacity is maximal at $\text{pH} = \text{pK}_a$** and useful only within $\text{pK}_a \pm 1$ — this is why the flat middle of a titration curve is the buffering zone.

$$\text{Henderson–Hasselbalch } \text{pH} = \text{pK}_a + \log \left(\frac{[\text{salt}]}{[\text{acid}]} \right) = \text{pK}_a + \log \left(\frac{[\text{A}^-]}{[\text{HA}]} \right)$$

$$\text{For blood: } \text{pH} = 6.1 + \log \left(\frac{\text{HCO}_3^-}{0.03 \times \text{pCO}_2} \right) = 6.1 + \log \left(\frac{24}{1.2} \right) = 6.1 + \log 20 = 7.4$$

The 20 : 1 ratio is the single most examined number in the unit. pH depends on the **ratio**, not absolute values — which is why compensation (renal HCO_3^- retention, respiratory CO_2 blow-off) restores pH by restoring the ratio.

BUFFER SYSTEM	pK_a	WHERE IT MATTERS
$\text{HCO}_3^- / \text{H}_2\text{CO}_3$	6.1	Chief ECF buffer — weak pK but an open system : lungs dump CO_2 , kidney regenerates HCO_3^-
Haemoglobin	~7.9 / 6.7	Chief intra-erythrocyte buffer; deoxy-Hb is the better buffer → isohydric CO_2 carriage
Protein (imidazole of His)	6.0	Main intracellular buffer; also plasma albumin
$\text{HPO}_4^{2-} / \text{H}_2\text{PO}_4^-$	6.8	Intracellular + urinary titratable acidity ; bone reservoir

Ammonia
($\text{NH}_3/\text{NH}_4^+$)

9.2

Renal tubular acid excretion in chronic acidosis

7.35–7.45

arterial blood pH

24

mM HCO_3^-

40

mmHg pCO_2

6.2–7.0

rumen pH (SARA < 5.6)

2–3

abomasum /
gastric

6.6–6.8

milk (↑ in mastitis)

Trap: herbivore urine is normally **alkaline** (7.5–8.5), carnivore urine **acidic** (5.5–7.0) — a "normal" pH 8 urine in a dog is abnormal (urease-positive infection).

06 DEFINITION, CLASSIFICATION & ISOMERISM

Carbohydrates = polyhydroxy aldehydes or ketones, or substances yielding them on hydrolysis; empirical formula $(CH_2O)_n$. Classified as **monosaccharides** (cannot be hydrolysed), **disaccharides**, **oligosaccharides** (2–10), **polysaccharides** (homo- vs hetero-).

Aldose –CHO (glucose)

Ketose C=O (fructose)

Epimers differ at ONE C: glucose ↔ galactose at **C-4**, glucose ↔ mannose at **C-2**

Anomers α/β at C-1 (anomeric carbon)

Enantiomers D/L — mirror at penultimate C; body sugars are **D**

Mutarotation α +112° → +52.7° → β +18.7°

- **Reducing sugars** have a free anomeric –OH → reduce Benedict's/Fehling's. **All monosaccharides**, maltose, isomaltose, lactose, cellobiose are reducing. **Sucrose and trehalose are non-reducing** (both anomeric carbons locked).
- **Ring forms:** pyranose (6-membered) & furanose (5-membered); Haworth projection; chair form of β-D-glucopyranose is the most stable sugar because **all bulky groups are equatorial**.
- **Derivatives:** sugar acids (gluconic, glucuronic — conjugation & detoxification, ascorbic acid), sugar alcohols (sorbitol → cataract in diabetes, glycerol, inositol), deoxy sugars (deoxyribose, fucose), **amino sugars** (glucosamine, galactosamine, **N-acetylneuraminic acid = sialic acid**), sugar esters (glucose-6-P).

07 MONOSACCHARIDES — BIOLOGICAL SIGNIFICANCE

SUGAR	WHY IT EXISTS IN THE ANIMAL
Ribose (deoxyribose)	Backbone of RNA / DNA , ATP, GTP, NAD ⁺ , FAD, CoA, SAM. Sole source = HMP shunt . Ribose-5-P is the entry point of nucleotide synthesis.
Glucose	"Blood sugar", universal fuel; the only fuel for brain, RBC, renal medulla under normal conditions. Stored as glycogen, converted to fat.
Fructose	Ketohexose, sweetest sugar ; enters glycolysis independent of insulin. Chief energy source of spermatozoa in bull/ram semen (fructolysis — a semen-quality test); high in fetal blood of ruminants.
Galactose	Made in the udder from glucose → lactose synthesis ; component of cerebrosides, gangliosides, glycoproteins, blood-group antigens. Galactosaemia if metabolism blocked.
Mannose	Core sugar of N-linked glycoproteins ; mannose-6-phosphate is the address tag that routes enzymes to lysosomes (I-cell disease when lost).

Amino sugars

Glucosamine & galactosamine → GAGs, cartilage, chitin; **sialic acid** caps glycoproteins & determines RBC lifespan and virus attachment (influenza).

Species hook: ruminant blood glucose is only **45–60 mg/dL** (dog/cat 70–110) because dietary carbohydrate is fermented to VFA in the rumen — the ruminant is a **gluconeogenic animal living on propionate**. This single fact explains ketosis, pregnancy toxemia and the whole of Unit 3.

08 DISACCHARIDES — LINKAGE IS THE ANSWER

SUGAR	UNITS & LINKAGE	RED.	NOTE
Maltose	Glc α1→4 Glc	Yes	Malt sugar; from starch by amylase
Isomaltose	Glc α1→6 Glc	Yes	From branch points of amylopectin
Lactose	Gal β1→4 Glc	Yes	Milk sugar , only in mammary gland; osmotic regulator of milk volume
Sucrose	Glc α1→2β Fru	No	Cane sugar; hydrolysis → invert sugar ; no free anomeric C
Cellobiose	Glc β1→4 Glc	Yes	From cellulose ; only rumen/hindgut microbes split it
Trehalose	Glc α1→1α Glc	No	"Insect blood sugar"; fungal

4.7%

lactose, cow milk

4.9%

buffalo

6.2%

mare — highest

≈0

lactase in adult cat/dog

Trap: lactose is the **only** sugar of purely animal origin. Adult ruminants and most adult carnivores lose intestinal lactase — feeding milk causes osmotic diarrhoea, a classic viva point.

09 POLYSACCHARIDES — HOMOLYCANS

POLYSACCHARIDE	MONOMER & LINKAGE	BRANCHING	VETERINARY SIGNIFICANCE
Starch — amylose	Glucose, α1→4	Unbranched helix	Stains deep blue with iodine; plant storage
Starch — amylopectin	Glucose, α1→4 + α1→6	Every 24–30 residues	Stains violet/purple ; 80 % of starch
Dextrins	Partial hydrolysis products	—	Erythroextrin (red with I ₂) → achroextrin (no colour); limit dextrin resists amylase
Dextran	Glucose, α1→6 chief	α1→3, α1→4	Bacterial; plasma volume expander (colloid) ; dental plaque matrix

Glycogen	Glucose, $\alpha 1 \rightarrow 4$ + $\alpha 1 \rightarrow 6$	Every 8–12 residues	Animal storage: liver 5–6 % (blood glucose homeostasis), muscle 0.7–1 % (local fuel only — no glucose-6-phosphatase). Dense branching = many non-reducing ends = rapid mobilisation. Stains red-brown with iodine.
Cellulose	Glucose, $\beta 1 \rightarrow 4$	Linear, H-bonded fibrils	No mammal makes cellulase; rumen/caecal microbes ferment it → VFA. Basis of herbivore nutrition & crude fibre.
Inulin	Fructose, $\beta 2 \rightarrow 1$	Linear	Freely filtered, not reabsorbed/secreted → gold-standard GFR marker ; prebiotic
Chitin	N-acetylglucosamine, $\beta 1 \rightarrow 4$	Linear	Arthropod exoskeleton & fungal cell wall — target of antifungals; second most abundant polysaccharide

One-line contrast: Starch, glycogen = α -linked = digestible by mammalian amylase. Cellulose, chitin = β -linked = need microbial enzymes. Digestibility is decided by the anomeric configuration, not by the sugar.

10 MUCOPOLYSACCHARIDES (GAGS) & BACTERIAL CELL-WALL POLYSACCHARIDES

GAGs = long unbranched heteropolysaccharides of a repeating disaccharide = **uronic acid (or galactose) + amino sugar**, almost all **sulphated** and strongly anionic. Bound to a core protein they form **proteoglycans** (ground substance) — a huge hydrated gel that resists compression, lubricates and traps cations and water.

GAG	REPEATING UNIT	SITE / USE
Hyaluronic acid	GlcUA + GlcNAc — only non-sulphated, no protein core	Synovial fluid, vitreous humour, Wharton's jelly, loose CT. Hyaluronidase = " spreading factor " of sperm acrosome & invasive bacteria
Chondroitin 4-/6-sulphate	GlcUA + GalNAc-SO ₄	Most abundant GAG — cartilage, bone, cornea; joint supplements
Keratan sulphate	Galactose + GlcNAc-SO ₄ (no uronic acid)	Cornea (transparency), cartilage, intervertebral disc
Dermatan sulphate	IdUA + GalNAc-SO ₄	Skin, blood vessels, heart valves
Heparin	IdUA-SO ₄ + GlcN-SO ₄	Most acidic / highest negative charge ; mast-cell granules; anticoagulant (antithrombin III); releases lipoprotein lipase — "clearing factor"
Heparan sulphate	Similar, less sulphated	Cell surface, basement membrane — glomerular charge barrier (loss → proteinuria)

Clinical: deficiency of lysosomal hydrolases → **mucopolysaccharidoses** (accumulation in cornea, bone, brain; reported in cats and dogs — MPS I, VI, VII). Corneal clouding + skeletal dysplasia + dwarfism is the classic triad.

Bacterial cell wall — peptidoglycan (murein). Alternating **N-acetylglucosamine (NAG)** and **N-acetylmuramic acid (NAM)** joined $\beta 1 \rightarrow 4$; NAM carries a tetrapeptide, and chains are cross-linked by **transpeptidase** (pentaglycine bridge in *Staphylococcus*).

NAG-NAM $\beta 1 \rightarrow 4$ → tetrapeptide → transpeptidase cross-link → rigid sacculus

- **Lysozyme** (tears, saliva, egg white, phagocytes) hydrolyses the $\beta 1 \rightarrow 4$ NAM-NAG bond → lysis. First-line innate defence.
- **Penicillins & cephalosporins** inhibit transpeptidase (PBP) → wall cannot cross-link. Selective toxicity rests on the fact that **the host has no peptidoglycan**.
- **Gram-positive:** thick peptidoglycan + **teichoic acid** (glycerol/ribitol phosphate — adhesion, antigenicity).
- **Gram-negative:** thin peptidoglycan + outer membrane bearing **lipopolysaccharide (LPS)**: O-antigen + core + **lipid A = the endotoxin** — the trigger of coliform mastitis, endotoxic shock and DIC (ties directly into Unit 3 "oxidative stress & shock").
- **Capsular polysaccharides** resist phagocytosis and are the basis of serotyping and of many vaccines.

11 BLOOR'S CLASSIFICATION — THE SKELETON OF EVERY LIPID ANSWER
LIPIDS (BLOOR)
SIMPLE — ester of FA + alcohol

- Fats & oils = triacylglycerol (FA + glycerol) — fat is solid, oil liquid at room temp
- Waxes = FA + long-chain alcohol

COMPOUND — + extra group

- Phospholipids (glycero- / sphingo-)
- Glycolipids (cerebroside, ganglioside)
- Lipoproteins (CM, VLDL, LDL, HDL)
- Sulpholipids, amino-lipids

DERIVED — hydrolysis products

- Fatty acids, glycerol, mono/di-AG
- Steroids & sterols (cholesterol)
- Ketone bodies, fat-soluble vit ADEK
- Eicosanoids, terpenes, carotenoids

Common properties: insoluble in water, soluble in organic solvents (ether, chloroform, benzene); **amphipathic** if a polar head exists; highest calorific value **9.3 kcal/g** (vs 4.1 for carbohydrate/protein); non-polar so stored anhydrous — the most economical fuel store.

Functions: concentrated energy, structural (membranes, myelin), thermal insulation & padding, carrier of vitamins ADEK, precursor of steroid hormones & eicosanoids, source of essential fatty acids, surfactant, milk-fat and egg-yolk quality.

12 FATTY ACIDS & ESSENTIAL FATTY ACIDS

- **Saturated:** butyric C4 (rumen VFA, milk fat), palmitic C16, stearic C18. **Unsaturated:** oleic 18:1 ω 9, **linoleic 18:2 ω 6**, **α -linolenic 18:3 ω 3**, arachidonic 20:4 ω 6, EPA 20:5, DHA 22:6.
- **Essential fatty acids = linoleic + α -linolenic** — mammals lack Δ 12 and Δ 15 desaturase so cannot insert double bonds beyond C-9. **Arachidonic acid is semi-essential** (made from linoleic).
- **EFA deficiency:** scaly dermatitis, dull dry coat, poor wound healing, impaired reproduction, fatty liver, $\uparrow$ transepidermal water loss.
- **Cat exception:** the cat has negligible Δ 6-desaturase $\rightarrow$ **arachidonic acid is dietary-essential for cats**, which is why cats are obligate carnivores nutritionally.
- **Natural cis** configuration bends the chain ($\uparrow$ fluidity, $\downarrow$ melting point); industrial **trans** behaves like saturated.

Trap: " ω " numbering counts from the **methyl** end, " Δ " from the **carboxyl** end. Arachidonic acid = 20:4 Δ 5,8,11,14 = ω 6.

13 FAT INDICES — PURE OBJECTIVE MARKS

INDEX	MEASURES	VALUE / USE
Saponification no.	mg KOH to saponify 1 g fat	Inversely $\propto$ chain length. Butter 210–230 (short chains), groundnut oil $\approx$ 190
Iodine no.	g I ₂ absorbed per 100 g fat	Degree of unsaturation. Coconut $\approx$ 8, butter 26–38, linseed 175–200
Acid no.	mg KOH to neutralise free FA in 1 g	Index of hydrolytic rancidity — the practical you perform
Reichert–Meissl no.	Volatile, water-soluble FA (butyric)	Butter 24–34; vanaspati <1 → test of ghee adulteration
Polenske no.	Volatile, water-insoluble FA	High in coconut oil — detects coconut-oil adulteration
Acetyl no.	Free –OH groups	High in castor oil (ricinoleic acid)

Rancidity. Hydrolytic = lipase/moisture → free FA, ↑acid number. **Oxidative** = O₂ attacks double bonds → peroxides, aldehydes; worst in ω -3-rich fish oil. Prevented by **vitamin E, BHT/BHA**. **Feed link:** rancid fish-oil diet + vitamin E deficiency → **yellow fat disease (steatitis / pansteatitis)** in cats and mink.

14 COMPOUND LIPIDS — THE ONES EXAMINERS NAME

LIPID	IDENTITY & ROLE
Lecithin (PC)	Choline; most abundant phospholipid; emulsifier ; dipalmitoyl-lecithin = lung surfactant ; egg yolk
Cephalin (PE)	Ethanolamine; brain; role in blood coagulation (thromboplastin)
Phosphatidyl-serine	Inner leaflet; externalised in apoptosis ; procoagulant surface
Phosphatidyl-inositol	PIP₂ → IP₃ + DAG : the Ca ²⁺ second-messenger system; GPI membrane anchors
Cardiolipin	Diphosphatidylglycerol — found only in the inner mitochondrial membrane ; antigenic (VDRL antigen)
Plasmalogens	Ether linkage at C-1; abundant in heart & muscle , myelin
Sphingomyelin	Only phospholipid with sphingosine, not glycerol ; myelin sheath; Niemann–Pick disease
Cerebrosides	Ceramide + one sugar (galactose); white matter; Gaucher/Krabbe
Gangliosides	Ceramide + oligosaccharide + sialic acid ; GM1 is the cholera-toxin receptor ; Tay–Sachs
Cholesterol	27-C, cyclopentanoperhydrophenanthrene ring, exclusively animal; precursor of bile acids, vitamin D₃, all steroid hormones ; membrane fluidity buffer

15 LIPOPROTEINS — DENSITY RISES AS FAT FALLS



CLASS	CARRIES	FROM → TO
Chylomicron	Exogenous TG (dietary)	Intestine → tissues; lymph/lacteals
VLDL	Endogenous TG	Liver → tissues; failure to export → fatty liver
LDL	Cholesterol ("bad")	→ peripheral tissue, receptor-mediated endocytosis
HDL	Cholesterol ("good")	Reverse cholesterol transport , tissue → liver; activates LCAT

- **Lipoprotein lipase** on capillary endothelium hydrolyses TG in CM/VLDL; activated by **apo C-II**, released into blood by **heparin** ("clearing factor lipase") — the reason post-prandial plasma is lipaemic/milky.
- **Species point:** dog, cat and most farm species are "**HDL animals**" (HDL predominant) → spontaneous atherosclerosis is rare; hypercholesterolaemia in dogs points to **hypothyroidism, diabetes, nephrotic syndrome, cholestasis or hyperadrenocorticism**.

16 PROSTAGLANDINS & THE EICOSANOID TREE

Eicosanoids = 20-carbon (*eicosa*) derivatives of **arachidonic acid**, released from membrane phospholipid by **phospholipase A₂**. They are **local hormones (autacoids)** — made on demand, act on the cell of origin or its neighbour, and are destroyed within seconds; they are **not stored**. Basic skeleton = prostanoic acid (cyclopentane ring + two side chains).

Membrane phospholipid → **phospholipase A₂** ⊥ **corticosteroids** → Arachidonic acid

└ **cyclo-oxygenase** ⊥ **NSAIDs / aspirin** → PGG₂ → PGH₂ → PGD/E/F, PGI₂, TXA₂

└ **lipoygenase** → HPETE → Leukotrienes (LTB₄, LTC₄/D₄/E₄ = SRS-A)

- **Nomenclature:** PG + letter (A–I, by ring substituent) + **subscript number = number of double bonds in the side chains**. Series 1 from dihomo-γ-linolenic, **series 2 from arachidonic (the important ones)**, series 3 from EPA — which is why fish-oil ω-3 gives **less inflammatory** eicosanoids.
- **Aspirin** irreversibly acetylates COX (platelets cannot resynthesise it → long antiplatelet effect); other **NSAIDs are reversible**; **corticosteroids block phospholipase A₂** and therefore switch off **both** arms — the reason steroids are broader anti-inflammatories than NSAIDs.

EICOSANOID	CHIEF SOURCE	PRINCIPAL ACTION
PGF_{2α}	Uterine endometrium	Luteolysis + myometrial contraction — the basis of oestrus synchronisation, induction of parturition, treatment of pyometra & retained placenta (dinoprost, cloprostenol)
PGE₂	Most tissues	Vasodilation, fever & pain sensitisation , gastric mucus/bicarbonate cytoprotection , bronchodilation, cervical relaxation
PGI₂ (prostacyclin)	Vascular endothelium	Vasodilation, inhibits platelet aggregation
TXA₂	Platelets	Vasoconstriction, promotes platelet aggregation — exact opposite of PGI ₂
LTB₄	Leucocytes	Chemotaxis of neutrophils
LTC₄/D₄/E₄	Mast cells	SRS-A : bronchoconstriction, ↑vascular permeability — anaphylaxis, equine asthma

The PGI₂ ↔ TXA₂ balance (endothelium vs platelet) governing haemostasis, and **PGF_{2α} luteolysis**, are the two prostaglandin facts most likely to be asked in both objective and long-answer form.

Trap: NSAID toxicity in dogs (gastric ulceration, renal papillary necrosis) is not a side reaction — it is the **predictable loss of PGE₂ cytoprotection and PGI₂-dependent renal perfusion**.

17 CLASSIFICATION OF PROTEINS

GROUP	MEMBERS & IDENTIFIERS
SIMPLE (yield only AA)	Albumin — water soluble, coagulable, full-saturation $(\text{NH}_4)_2\text{SO}_4$; serum albumin, ovalbumin, lactalbumin. Globulin — insoluble in water, soluble in dilute salt, half-saturation salting-out; immunoglobulins. Glutelin (glutenin), Prolamin (zein of maize, gliadin — deficient in lysine & tryptophan). Histone — basic, arginine/lysine rich, bound to DNA. Protamine — most basic, in fish sperm. Albuminoid / scleroprotein — most insoluble: collagen, elastin, keratin.
CONJUGATED (+ prosthetic group)	Nucleoprotein (nucleic acid) · Glycoprotein / mucoprotein (CHO — mucin, hormones) · Phosphoprotein (casein, ovovitellin) · Chromoprotein (haem — Hb, myoglobin, cytochromes, catalase) · Lipoprotein · Metalloprotein (ceruloplasmin—Cu, ferritin—Fe, carbonic anhydrase—Zn) · Flavoprotein (FAD).
DERIVED	Primary (bonds intact): proteans, metaproteins, coagulated proteins. Secondary (progressive hydrolysis): protein → proteose → peptone → polypeptide → peptide → amino acid.

Functional list for a 10-mark "biological significance" answer: structural (collagen, keratin) · catalytic (enzymes) · transport (Hb, albumin, transferrin, ceruloplasmin) · hormonal (insulin, GH) · defence (immunoglobulins, complement) · clotting (fibrinogen) · contractile (actin, myosin) · **colloid osmotic pressure (albumin ≈ 80 %)** · buffering · storage (casein, ferritin, ovalbumin) · genetic (histones) · nutrition (10 essential AA) · receptors & signalling.

18 FOUR LEVELS OF STRUCTURE

Primary	Linear amino-acid sequence + disulphide positions. Bond: peptide (covalent) . Determines all higher orders. First sequenced protein = insulin (Sanger) . A single substitution can be lethal (Hb-S).
Secondary	Local regular folding held by hydrogen bonds between backbone C=O and N-H. α-helix : right-handed, 3.6 residues per turn, pitch 5.4 Å , H-bond $n \rightarrow n+4$; broken by proline ("helix breaker") & charged clusters. β-pleated sheet : parallel/antiparallel, H-bonds between strands (silk fibroin, keratin). Collagen triple helix : left-handed strands wound right, repeating Gly-X-Y (X often proline, Y hydroxyproline); glycine at every third residue is obligatory.

Tertiary

Overall 3-D shape of one chain; stabilised chiefly by **hydrophobic interactions**, plus ionic/salt bridges, H-bonds, van der Waals and **disulphide bonds** (the only covalent one). Creates the active site / binding cleft. Example: myoglobin.

Quaternary

Association of ≥ 2 subunits by non-covalent forces. **Haemoglobin $\alpha_2\beta_2$** , LDH, aldolase, immunoglobulins. Basis of **co-operativity and allosteric regulation** (sigmoid O_2 curve of Hb vs hyperbolic curve of myoglobin).

Vitamin C link: ascorbate is the cofactor for **prolyl and lysyl hydroxylase**. No vitamin C $\rightarrow$ no hydroxyproline $\rightarrow$ collagen triple helix cannot cross-link $\rightarrow$ **scurvy** (guinea pig, primates, fish — species that lack L-gulonolactone oxidase). Poor wound healing + capillary fragility.

19 PROPERTIES OF PROTEINS

- **Amphoteric & isoelectric point (pI):** net charge zero at pI $\rightarrow$ minimum solubility, maximum precipitation, no movement in an electric field. **Casein pI = 4.6** — the reason milk curdles when it sours or when acid/rennet is added, and the basis of paneer/dahi. Serum albumin pI ≈ 4.9 .
- **Electrophoresis:** at pH 8.6 all plasma proteins are anionic and separate as **albumin $\rightarrow \alpha_1 \rightarrow \alpha_2 \rightarrow \beta \rightarrow \gamma$** . This is the direct link to **dysproteinemias** in Unit 3.
- **Denaturation:** heat, extremes of pH, urea, guanidine, heavy metals, organic solvents, UV, vigorous shaking. **Secondary, tertiary and quaternary structure are destroyed; the peptide bonds of the primary structure remain intact.** Result: loss of biological activity, $\downarrow$ solubility, $\uparrow$ viscosity, $\uparrow$ digestibility, exposure of $-SH$ groups. Usually irreversible (boiled egg) but sometimes reversible (ribonuclease).
- **Precipitation:** by heavy-metal ions (Hg^{2+} , Pb^{2+} — protein/egg white is a first-aid antidote), by alkaloidal reagents (tungstic, trichloroacetic, picric, sulphosalicylic acid — used for deproteinising blood before glucose estimation), by organic solvents, and by **salting out** — globulin at half saturation and albumin at full saturation with ammonium sulphate.
- **Colloidal nature:** large size, Tyndall effect, non-dialysable, exert osmotic (oncotic) pressure ≈ 25 mmHg.

20 AMINO ACIDS — STRUCTURE & CLASSIFICATION

General formula: a carbon bearing $-NH_2$, $-COOH$, $-H$ and a distinguishing **R group**. All 20 protein amino acids are **α -amino acids of the L-configuration; glycine alone is optically inactive** ($R = H$, no chiral carbon).

BASIS	GROUPS
Side chain	Aliphatic (Gly, Ala, Val, Leu, Ile) · Hydroxyl (Ser, Thr, Tyr) · Sulphur (Cys, Met) · Aromatic (Phe, Tyr, Trp — absorb at 280 nm; basis of protein estimation) · Acidic (Asp, Glu) + amides (Asn, Gln) · Basic (Lys, Arg, His) · Imino (Pro — actually an imino acid, gives yellow with ninhydrin)

Polarity	Non-polar/hydrophobic (buried in core) vs polar uncharged vs charged
Nutrition	Essential (10) · Semi-essential: arginine & histidine (needed in growth) · Non-essential
Metabolic fate	Purely ketogenic: Leucine & Lysine only. Both keto- & glucogenic: Phe, Tyr, Trp, Ile, Thr. All the rest glucogenic.

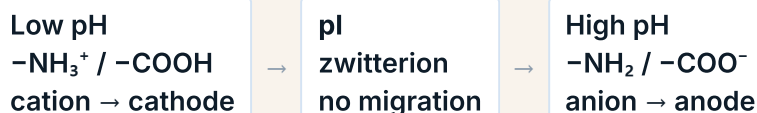
Essential amino acids — PVT TIM HALL: Phenylalanine, Valine, Threonine, Tryptophan, Isoleucine, Methionine, Histidine, Arginine, Leucine, Lysine.

Ketogenic — "Leu & Lys are Kings": only these two.

Species & feed hooks: **lysine** is first-limiting in cereal-based pig diets; **methionine** first-limiting in poultry; **tryptophan** deficient in maize (pellagra, since Trp → niacin). **Glycine/serine is essential for poultry** (uric acid synthesis). **Taurine is essential for the cat** — deficiency causes **dilated cardiomyopathy and feline central retinal degeneration**; **arginine deficiency in the cat causes hyperammonaemia within hours**. Rumen microbes synthesise all EAA, so quality of dietary protein matters far less in adult ruminants (but **bypass/rumen-protected methionine & lysine** raise milk protein).

21 PHYSICAL & CHEMICAL PROPERTIES OF AMINO ACIDS · THE PEPTIDE BOND

Amphoteric nature & zwitterion. In solution an amino acid exists as a dipolar ion with $-\text{NH}_3^+$ and $-\text{COO}^-$. It therefore acts as an acid **and** a base, and is a good buffer. Hence high melting points, water solubility, and large dipole moments — they behave like salts, not organic solids.



Isoelectric pH: for a neutral amino acid $\text{pI} = (\text{pK}_1 + \text{pK}_2)/2$; for acidic and basic amino acids, average the two pK values that flank the zwitterion. Solubility and buffering are **minimum** at pI.

Optical activity. The α -carbon is asymmetric (except glycine) → D and L forms, dextro (+) or laevo (−) rotation. **Proteins contain only L-amino acids**; D-forms occur in bacterial cell walls (D-alanine, D-glutamate) and some antibiotics — which is why they resist host proteases.

Peptide bond formation. The $-\text{COOH}$ of one amino acid condenses with the $-\text{NH}_2$ of the next with loss of water, giving a **substituted amide ($-\text{CO}-\text{NH}-$)** link. Chains are written N-terminal → C-terminal.

- Has **partial double-bond character** from resonance → the six atoms of the bond are **rigid and planar**; **no free rotation about the C–N bond**.
- Almost always **trans** (except X-proline); rotation is possible only about the ϕ (**phi**) and ψ (**psi**) bonds of the α -carbon — the basis of the **Ramachandran plot**.
- Uncharged but polar; the C=O and N–H are exactly the H-bond donors/acceptors that build the α -helix and β -sheet.

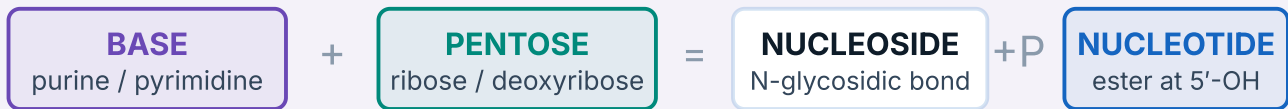
- Hydrolysed by strong acid/alkali or by proteolytic enzymes; two or more peptide bonds give the **biuret reaction**.

Trap: Biuret is negative for free amino acids and for dipeptides — it needs at least two peptide bonds (i.e. a tripeptide onwards). Ninhydrin, in contrast, is positive for free α -amino acids.

REACTION	DETECTS	COLOUR
Ninhydrin	Free α -amino acids	Blue-violet (Ruhemann's purple); proline → yellow
Biuret	≥ 2 peptide bonds	Violet / pink
Millon's	Tyrosine (phenol)	Brick red
Xanthoproteic	Aromatic ring	Yellow → orange with alkali
Hopkins–Cole	Tryptophan (indole)	Violet ring
Sakaguchi	Arginine (guanidino)	Red
Lead acetate	Cysteine (–SH)	Black PbS
Sanger's (FDNB)	N-terminal AA	Yellow DNP-AA
Edman (PITC)	N-terminal, sequential	PTH-amino acid

Sørensen's formol titration (your practical): formaldehyde blocks the $-\text{NH}_2$ group, removing its basic character, so the $-\text{COOH}$ can be titrated against standard NaOH with phenolphthalein — this is how amino acids are **estimated**, and it is a favourite "principle" question.

22 BASES → NUCLEOSIDE → NUCLEOTIDE



N-9 of purine or N-1 of pyrimidine joins C-1' of the sugar · β -configuration

Purines	Fused 6+5 ring. Adenine, Guanine ; also hypoxanthine, xanthine, uric acid (end product in birds & reptiles = uricotelic; in Dalmatian dogs), and methylxanthines caffeine & theobromine — theobromine of chocolate is the classic canine toxicosis .
Pyrimidines	Single 6-ring. Cytosine (both), Thymine (DNA only), Uracil (RNA only) ; orotic acid is the pathway intermediate. Mnemonic: CUT the PY — Cytosine, Uracil, Thymine are PYrimidines.
Naming	Adenine→adenosine→AMP/ADP/ATP; Guanine→guanosine→GMP; Cytosine→cytidine; Uracil→uridine; Thymine→thymidine. Add "deoxy" for the DNA series.

Biological significance of nucleotides (a guaranteed short note): ① **energy currency** — ATP, GTP, and the high-energy phosphoanhydride bond; ② **second messengers** — cAMP, cGMP; ③ **coenzymes** — NAD⁺, NADP⁺, FAD, CoA, all built on an **adenine nucleotide**; ④ **activated carriers** — UDP-glucose (glycogen), CDP-choline (phospholipid), **SAM** (methyl), PAPS (sulphate, for GAGs & detoxification); ⑤ **monomers** of DNA & RNA; ⑥ allosteric regulators (ATP/AMP ratio); ⑦ drug targets — allopurinol, 5-fluorouracil, azathioprine.

23 DNA — WATSON & CRICK, 1953

- Two antiparallel polynucleotide chains** (one 5'→3', the other 3'→5') wound into a **right-handed double helix**; sugar-phosphate backbone outside joined by **3',5'-phosphodiester bonds**, bases stacked inside.
- Complementary base pairing: A = T (2 hydrogen bonds), G ≡ C (3 hydrogen bonds)** — always a purine with a pyrimidine, keeping the diameter constant.
- Chargaff's rules:** A = T, G = C, therefore purines = pyrimidines; the (A+T)/(G+C) ratio is species-specific.
- Major and minor grooves — the binding sites of regulatory proteins and many drugs.

2 nm
diameter (20 Å)

10
bp per turn

3.4 nm
pitch per turn

0.34 nm
rise per bp

- Forms: B-DNA** = physiological, right-handed; **A-DNA** = dehydrated, right-handed; **Z-DNA** = **left-handed**, zig-zag, alternating purine-pyrimidine (GC repeats).

- **Denaturation / melting:** heat or alkali separates the strands. T_m rises with **G+C content** (3 H-bonds). Strand separation exposes the bases → **hyperchromic effect: absorbance at 260 nm increases**. Slow cooling → renaturation/annealing (the principle of hybridisation and PCR).
- **Packaging:** DNA + **histones (H2A, H2B, H3, H4 octamer)** wrapped by ~146 bp = **nucleosome**, sealed by **H1** → solenoid → chromatin → chromosome. Supercoiling is managed by **topoisomerases/gyrase** (target of fluoroquinolones in bacteria).
- **Functions:** repository of genetic information, replication (semi-conservative), transcription, mutation & variation, molecular diagnosis (PCR), parentage & breed identification.

24 RNA — A TYPICAL RIBONUCLEIC ACID

Single-stranded, **ribose** sugar, **uracil in place of thymine**, alkali-labile (2'-OH), can fold on itself into local double-helical hairpins.

TYPE	%	STRUCTURE & ROLE
mRNA	2–5	Most heterogeneous, shortest-lived. Eukaryotic mRNA is capped with 7-methylguanosine at 5' and carries a 3' poly-A tail ; carries codons — the working copy of the gene
tRNA	≈15	Smallest RNA ; clover-leaf secondary / inverted-L tertiary shape; has an anticodon arm and a 3'-CCA end that accepts the amino acid ; rich in unusual bases (pseudouridine, dihydrouridine, inosine). Adaptor molecule
rRNA	≈80	Most abundant ; with proteins forms the ribosome — prokaryote 70S (50S + 30S), eukaryote 80S (60S + 40S) ; the peptidyl transferase is a ribozyme . Target of aminoglycosides & macrolides
Others	—	snRNA (splicing), miRNA/siRNA (gene silencing), hnRNA (mRNA precursor)

Trap that costs marks: "smallest" = **tRNA**; "most abundant" = **rRNA**; "most variable / shortest half-life" = **mRNA**. Examiners rotate these three every year.

25 FINAL RECALL BANK — READ THESE LAST, STRAIGHT BEFORE THE PAPER

Fluid mosaic model proposed by	Singer & Nicolson, 1972
Membrane with highest lipid / lowest protein	Myelin
Membrane with highest protein	Inner mitochondrial membrane
Enzyme moving lipid inward / outward	Flippase / floppase
Only insulin-dependent glucose transporter	GLUT-4
Glucose sensor of β -cell & liver	GLUT-2 (high K_m)
Inhibitor of Na^+/K^+ -ATPase	Ouabain / digitalis
Na^+ -glucose symporter inhibitor	Phlorizin

Donnan equation	$[Na]_i[Cl]_i = [Na]_o[Cl]_o$
Buffer works best at	$pH = pK_a (\pm 1)$
pK_a of bicarbonate system	6.1
$HCO_3^- : H_2CO_3$ ratio in blood	20 : 1
Chief intracellular buffer	Protein (imidazole) & phosphate
Chief buffer inside the RBC	Haemoglobin
Epimers of glucose at C-4 / C-2	Galactose / mannose
Specific rotation of D-glucose	+52.7°
Sweetest sugar	Fructose
Energy source of spermatozoa	Fructose
Non-reducing disaccharides	Sucrose, trehalose
Milk sugar & its linkage	Lactose, Gal $\beta 1 \rightarrow 4$ Glc
Glycogen branch frequency	Every 8–12 residues ($\alpha 1 \rightarrow 6$)
Polysaccharide used to measure GFR	Inulin
Only non-sulphated GAG	Hyaluronic acid
Most acidic GAG / anticoagulant	Heparin
"Spreading factor" enzyme	Hyaluronidase
Bacterial wall polymer & its sugars	Peptidoglycan — NAG + NAM
Enzyme in tears that lyses bacteria	Lysozyme ($\beta 1 \rightarrow 4$ NAM–NAG)
Endotoxin of Gram-negative wall	Lipid A of LPS

Energy value of fat	9.3 kcal/g
Essential fatty acids	Linoleic & α -linolenic
EFA additionally essential for the cat	Arachidonic acid
Index of unsaturation	Iodine number
Index of rancidity	Acid number
Test for ghee adulteration	Reichert–Meissl number
Saponification number is $\propto$	1 / fatty-acid chain length
Phospholipid of lung surfactant	Dipalmitoyl lecithin
Phospholipid only in inner mito membrane	Cardiolipin
Only phospholipid without glycerol	Sphingomyelin
Largest, least dense lipoprotein	Chylomicron

Reverse cholesterol transport	HDL (apo A-I, LCAT)
Activator of lipoprotein lipase	Apo C-II (released by heparin)
Ring system of cholesterol	Cyclopentanoperhydrophenanthrene
Precursor of all eicosanoids	Arachidonic acid (20 C)
Enzyme blocked by corticosteroids	Phospholipase A ₂
Enzyme blocked by aspirin/NSAIDs	Cyclo-oxygenase
Prostaglandin causing luteolysis	PGF _{2α}
Platelet aggregator vs inhibitor	TXA ₂ vs PGI ₂
Most basic protein	Protamine
Phosphoprotein of milk & its pl	Casein, pl 4.6
Residues per turn of α-helix	3.6 (pitch 5.4 Å)
Repeating unit of collagen	Gly-X-Y (Y = hydroxyproline)
Bond retained after denaturation	Peptide bond (primary structure)
Only optically inactive amino acid	Glycine
Purely ketogenic amino acids	Leucine & lysine
Amino acids absorbing at 280 nm	Phe, Tyr, Trp
Ninhydrin colour with proline	Yellow (others blue-violet)
Method for estimating amino acids	Sørensen's formol titration
Hydrogen bonds in A=T / G≡C	2 / 3
Base pairs per turn & pitch of B-DNA	10 bp, 3.4 nm
Left-handed DNA	Z-DNA
Increase in A ₂₆₀ on melting	Hyperchromic effect
Smallest / most abundant RNA	tRNA / rRNA
Amino-acid attachment site on tRNA	3'-CCA end

Thirty-second closing sweep: 1972 Singer-Nicolson · 3 Na out : 2 K in · [Na][Cl] products equal · pH = pK_a + log salt/acid, ratio 20:1 · α = digestible, β = microbial · branch every 8–12 in glycogen · hyaluronic acid alone unsulphated · iodine = unsaturation, acid = rancidity, RM = ghee · PLA₂ → COX → PGF_{2α} lyses the CL · 3.6 residues per helical turn · PVT TIM HALL · Leu & Lys are the only ketogenic pair · A=T two bonds, G≡C three · 10 bp, 3.4 nm, right-handed B-DNA.